

VIVEK JOSH Y

✉ contact@vivekjoshy.com • github.com/vivekjoshy

OPENSkill

Bayesian Multiplayer Rating Library

2024

Open-source Bayesian rating library for asymmetric multi-team, multiplayer matches, implementing the Weng–Lin models with extensions. Published in the Journal of Open Source Software; preprint on arXiv since January 2024. Roughly 112,000 downloads a month, independently ported to eight languages, and cited 11 times including in IEEE Transactions on Games, COLING, and the Journal of the Royal Statistical Society. Pinned as a dependency in Ray/RLlib and the rating layer in Neural MMO; in production use at OpenAI, MultiVersus, Hunt: Showdown, and BeyondAllReason — and the rating engine behind the Vairified platform.

openskill.me • github.com/vivekjoshy/openskill.py • DOI: 10.21105/joss.05901 • arXiv:2401.05451

PROFESSIONAL EXPERIENCE

Vairified Corp

January 2025 – Present

Lead Scientist, Data & ML

Naples, Florida

- Sole data scientist; one of three in technical leadership.
- Own the backend ML platform — a 140k+ LOC NestJS API, with the Partner API shipped to production behind OAuth and scoped API keys, plus official Python and TypeScript SDKs.
- Built VAIR, the production rating engine: Thurstone–Mosteller via OpenSkill, score-margin weighting, adaptive uncertainty, and full test coverage.
- Rebuilt Sentinel, which AST-extracts roughly 3,000 business rules across three repositories and presents them in plain English for stakeholder review.
- Led the legacy PHP migration: 184 files and 39 screens mapped across a 119-action cutover.
- Delivered multisport sport-filtering end to end, from schema through cross-sport search to per-sport rating buckets.

Pickleball Players Network

October 2023 – December 2024

Data Scientist

Los Angeles, California

- Created matchmaking algorithms and skill assessment models.
- Built prototype system using Python and PostgreSQL.

Open Source Development

2016 – Present

- Maintain OpenSkill, a peer-reviewed Bayesian rating library with production deployments across games and sports.
- Build and operate supporting infrastructure: CI/CD pipelines, packaging, and documentation sites.

RESEARCH

Boundary Kernel Theory & Grammar Induction

In progress

Monograph in progress; the line of work runs back to OpenGrammar (2022). Open quantities: k^* , the interaction order, and the renormalizability conjectures.

Glia — Glass-Box Defeasible-Inheritance Reasoner

In progress

Settles inferences by graded biased competition rather than next-token prediction, emitting a derivation trace for every conclusion. Exercised against a five-topology stress battery and a 254k-statement run over GenericsKB.

ARC-AGI

Ongoing

Two years across roughly twelve solver architectures. Finding: the binding constraint is object-level relational structure that is not recoverable from pixel context — not compute.

TECHNICAL PROJECTS

NeurIPS 2024 — Predict New Medicines with BELKA	2024
• Silver medal: 61st of 1,950 teams. • Implemented domain adversarial training with gradient reversal. • Designed a custom molecular data processing architecture.	
SAT/SMT Solver	2024
• Built CDCL -based reinforcement learning system for SAT/SMT problems. • Custom rewards created for efficient solution space exploration.	
Prompt Ensemble Framework	2024
• Built an entropy -based framework for token evidence aggregation. • Created logarithmic pooling for confidence scoring.	

COMMUNITY

Machine Learning Street Talk	Present
Expert contributor on the MLST community server.	
The Hitchhiker's Guide to Python	2017
Contributed to Common Gotchas chapter (docs.python-guide.org)	

SKILLS

Technologies: PyTorch, Polars/Pandas, FastAPI, Ray/RLlib, TypeScript, NestJS

DevOps & Data: Git, Docker, AWS, GCP, PostgreSQL, MongoDB, TypeDB, RDF/SPARQL

Core: Formal Methods, Grammar Induction, Neuro - Symbolic Computation, Interpretable AI, Bayesian Inference

EDUCATION

Mahatma Gandhi University	2024
<i>Bachelor of Science in Bioinformatics</i>	Aluva, India